

ECEN 474/704: Lab 03

Layout Design Techniques (Layout Techniques for Current Mirror,Capacitor and Resistor)

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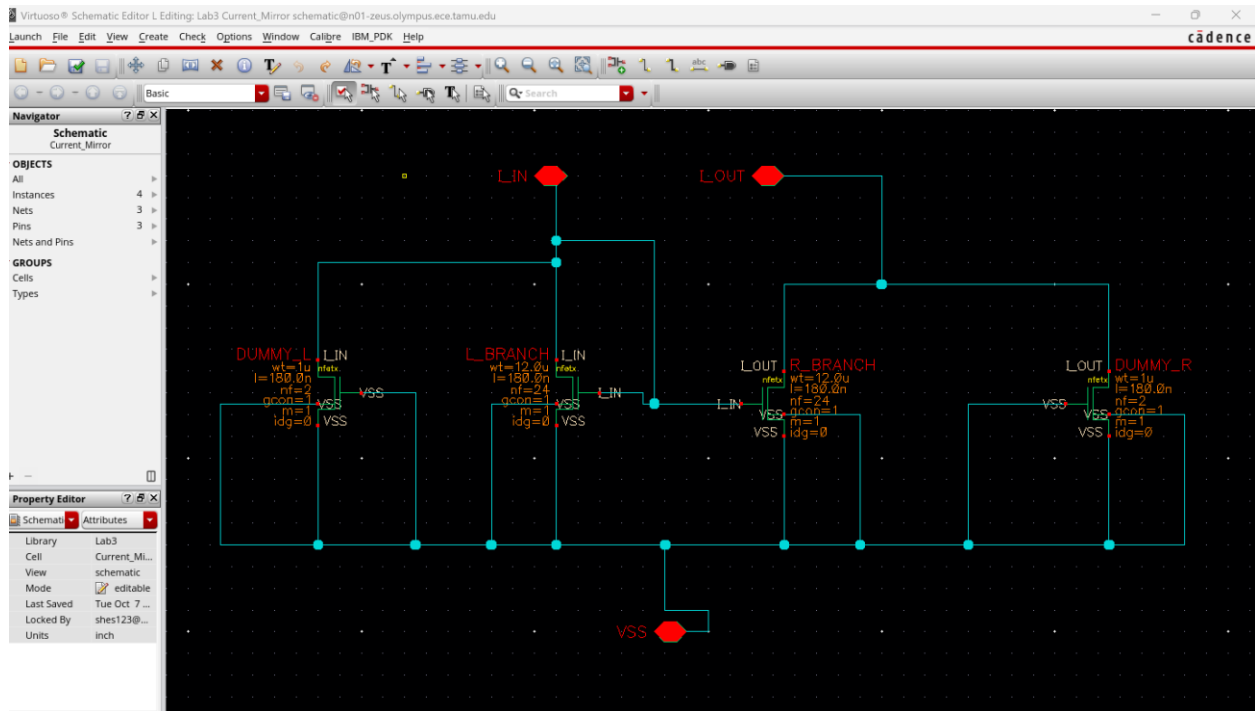
Due date: 2025-10-10

I. DISCUSSION OF THE LAB AND RESULTS

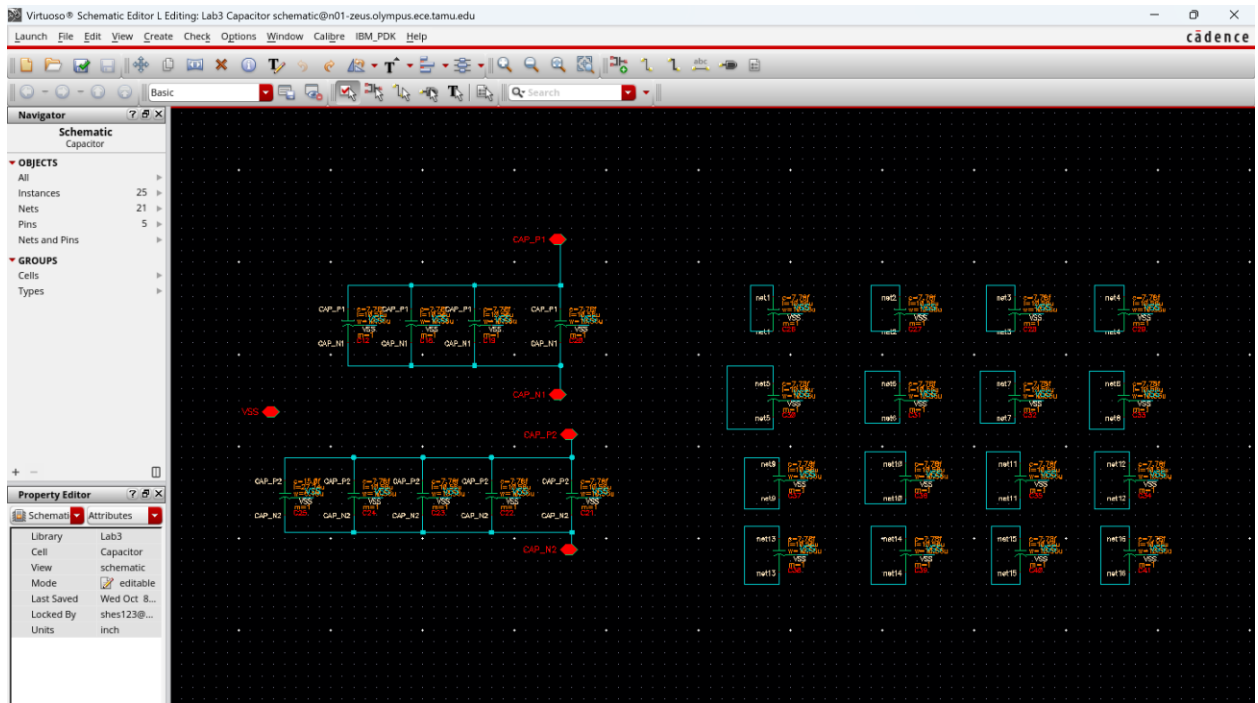
In this lab we studied about the layout technique of common centroid technique for Current Mirror,capacitor and resistor. In the pre-lab simulation tools were not used instead we relied on calculation to find the dimensions of the non-unit capacitances of C2 ,also in case of resistor the length and width corresponding to the area was calculated. In this lab using cadence virtuoso software the circuits were drawn in schematic and also the layout is laid,parallely the DRC and LVS is also run. The floor plan is also was created using power point tool. The lab is focused on fixing the drc errors and coming out with an effective layout design strategies.

II. SUB-CIRCUIT AND TEST BENCH SCHEMATICS

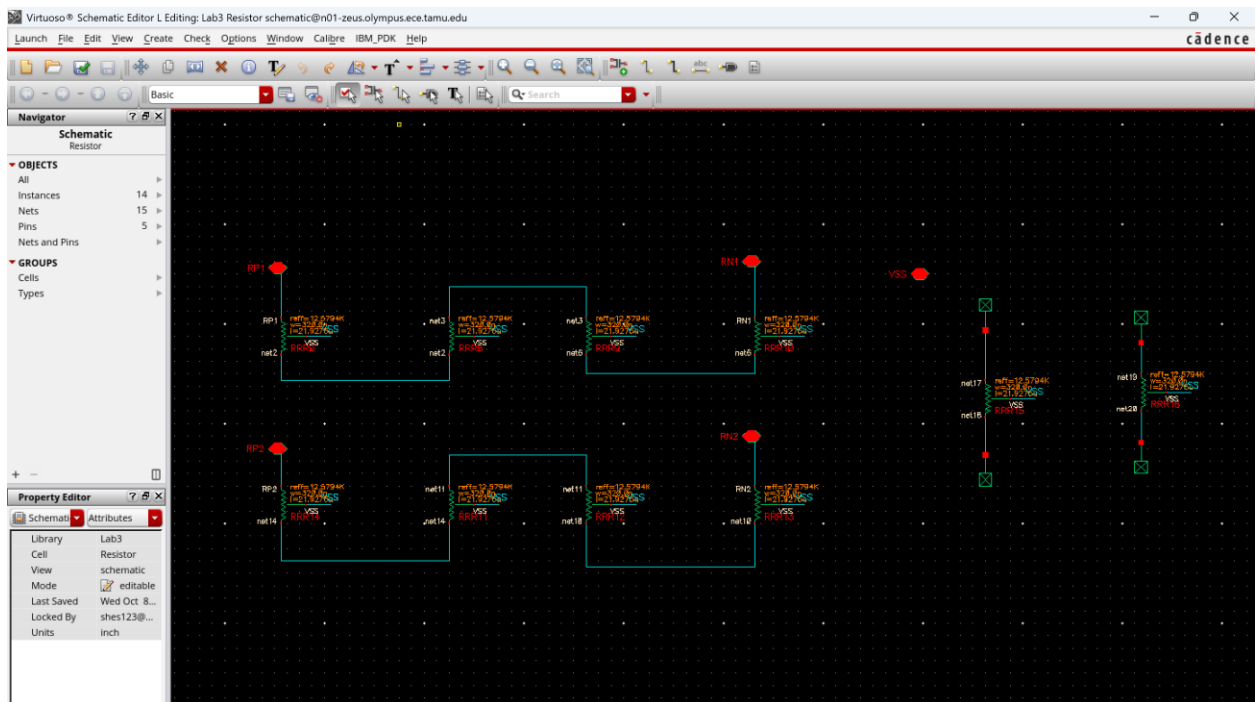
1. Current Mirror Schematic



2. Capacitor Schematic

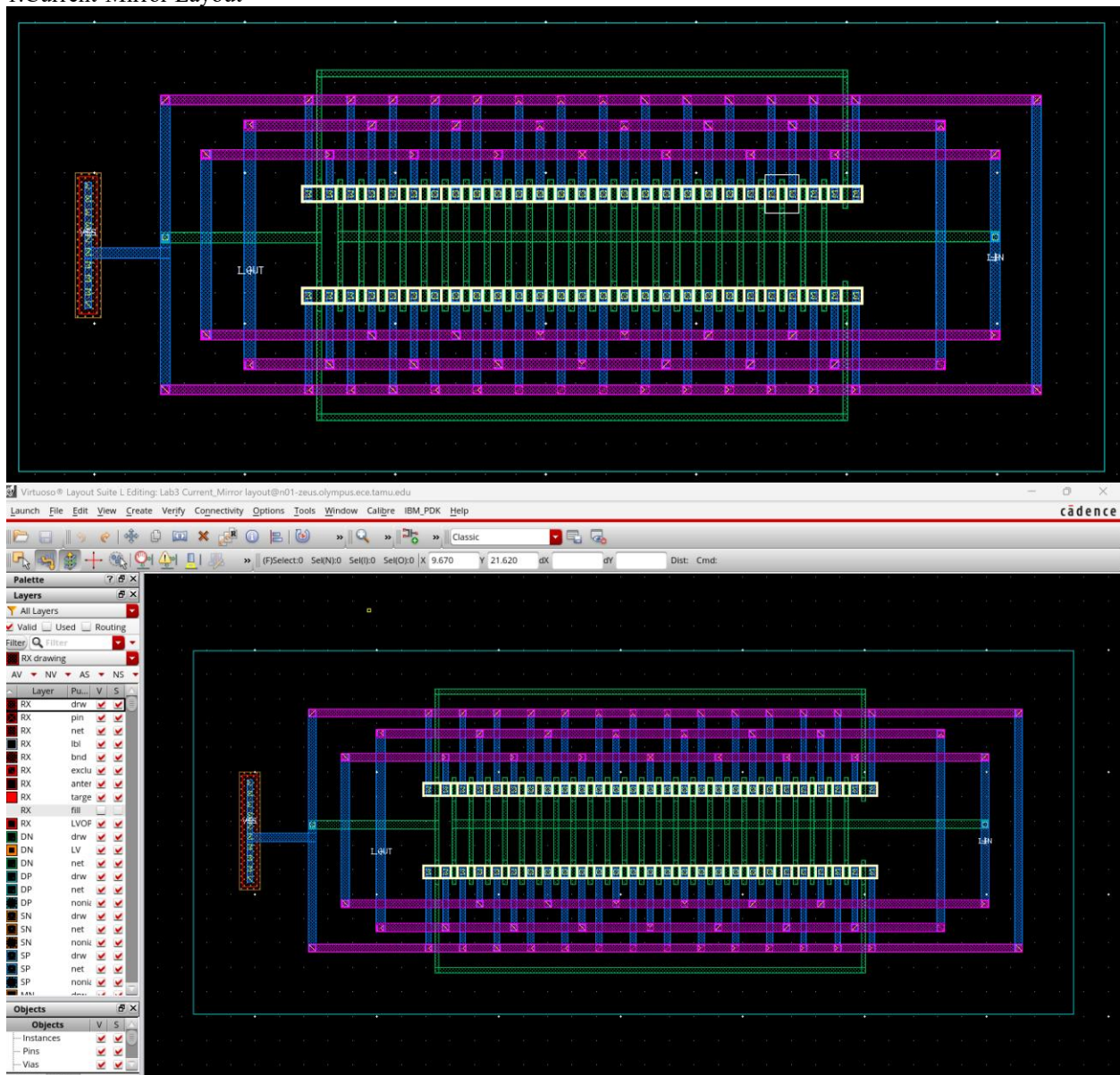


2.Resistor Schematic

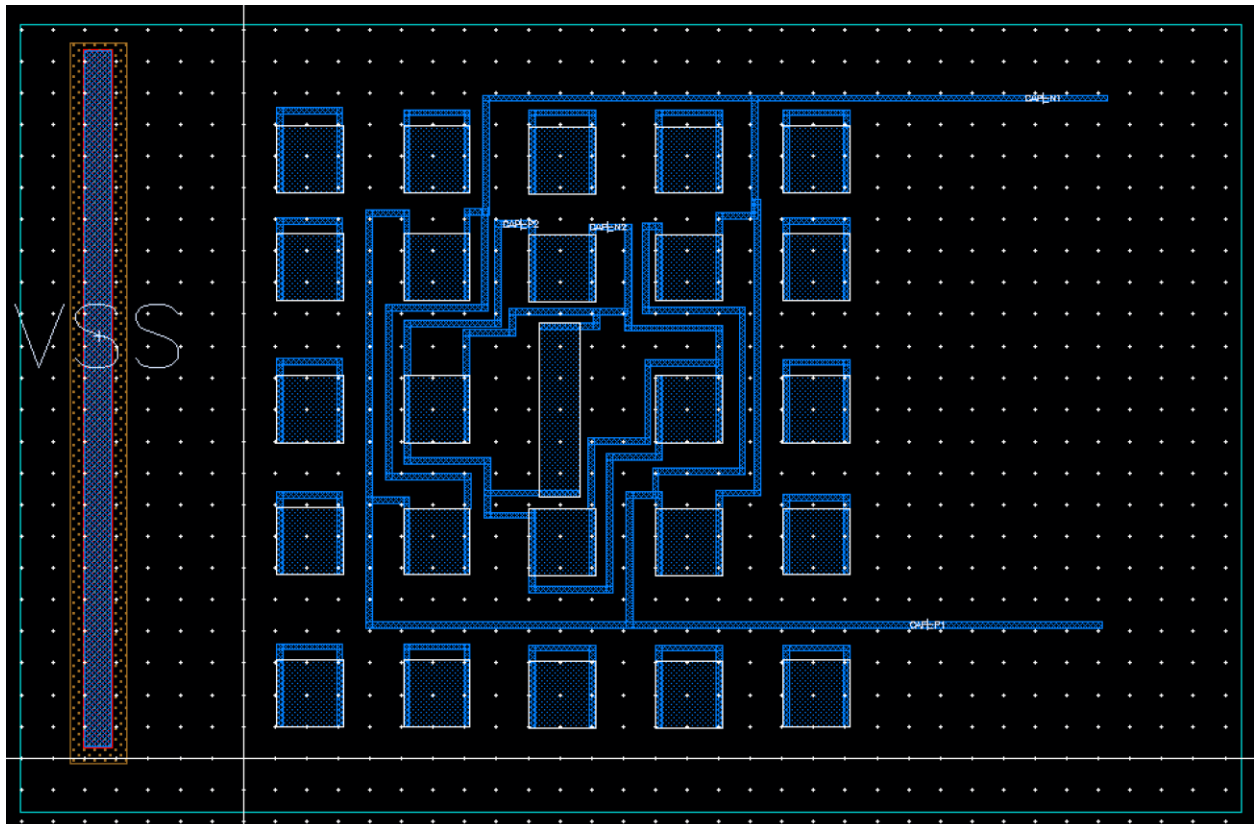


III. LAYOUT(S) AND FLOOR PLAN(S)

1.Current-Mirror Layout

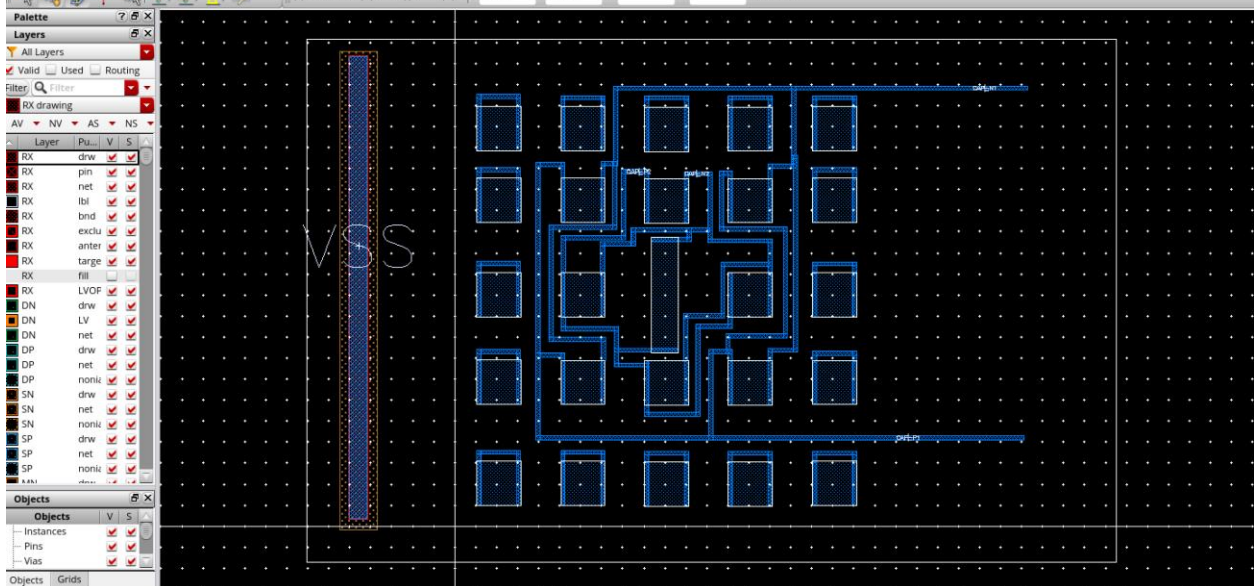


2.Capacitor Layout

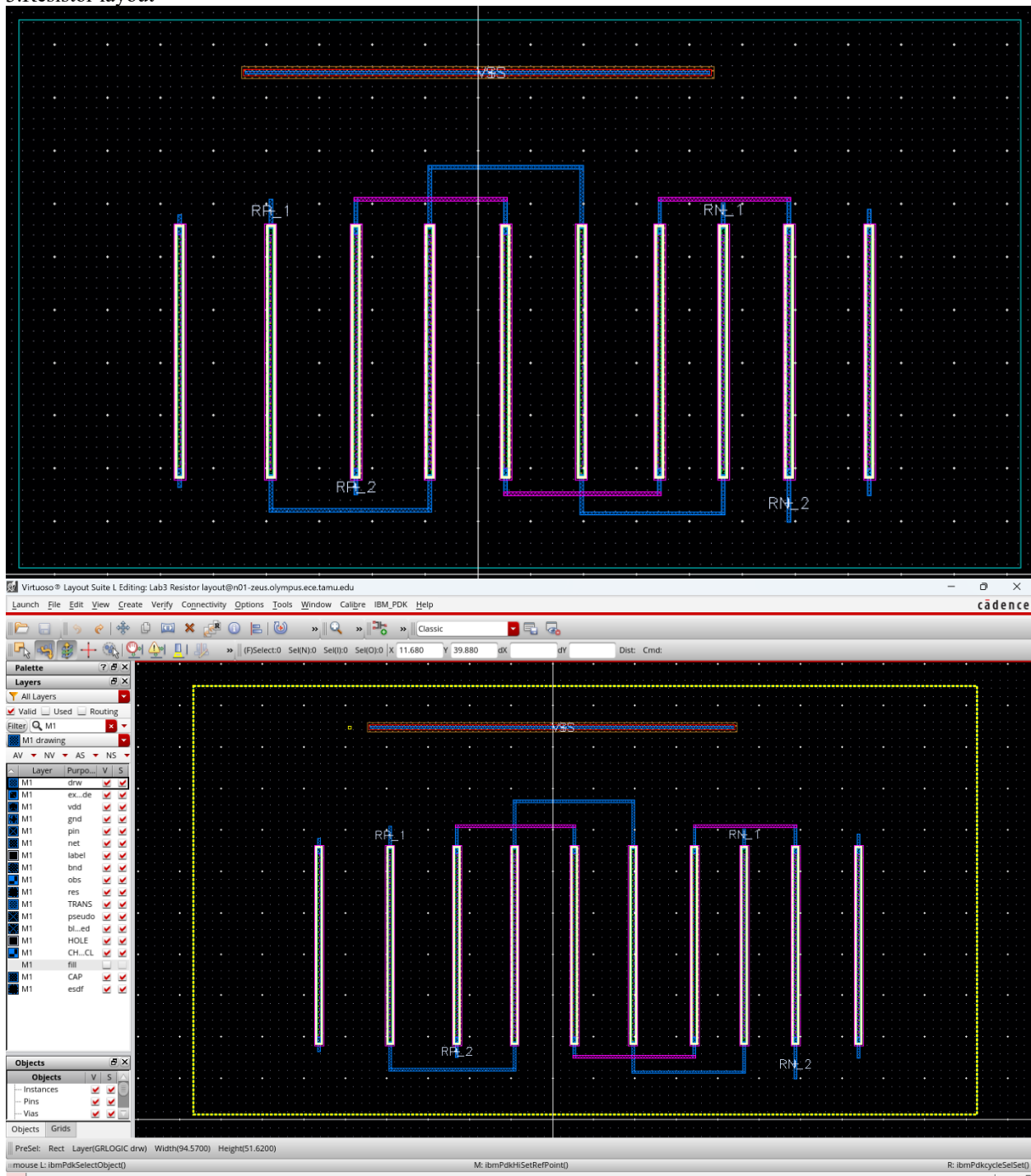


Virtuoso® Layout Suite L Editing: Lab3 Capacitor layout@n01-zeus.olympus.ece.tamu.edu
 Launch File Edit View Create Verify Connectivity Options Tools Window Calibre IBM_PDK Help

Classic
 (F)Select:1 Sel(N):0 Sel(I):0 Sel(O):1 X 75.600 Y 54.940 dX dY Dist: Cmd



3.Resistor layout

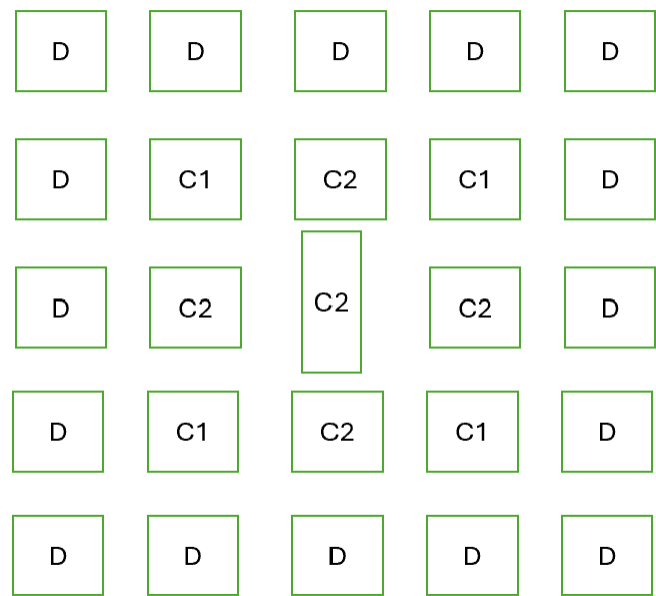


Floor Plan: (DUMMY REPRESENTED AS D INSTEAD OF *)

1.Current Mirror Floor Plan:



2.Capacitor Floor Plan:

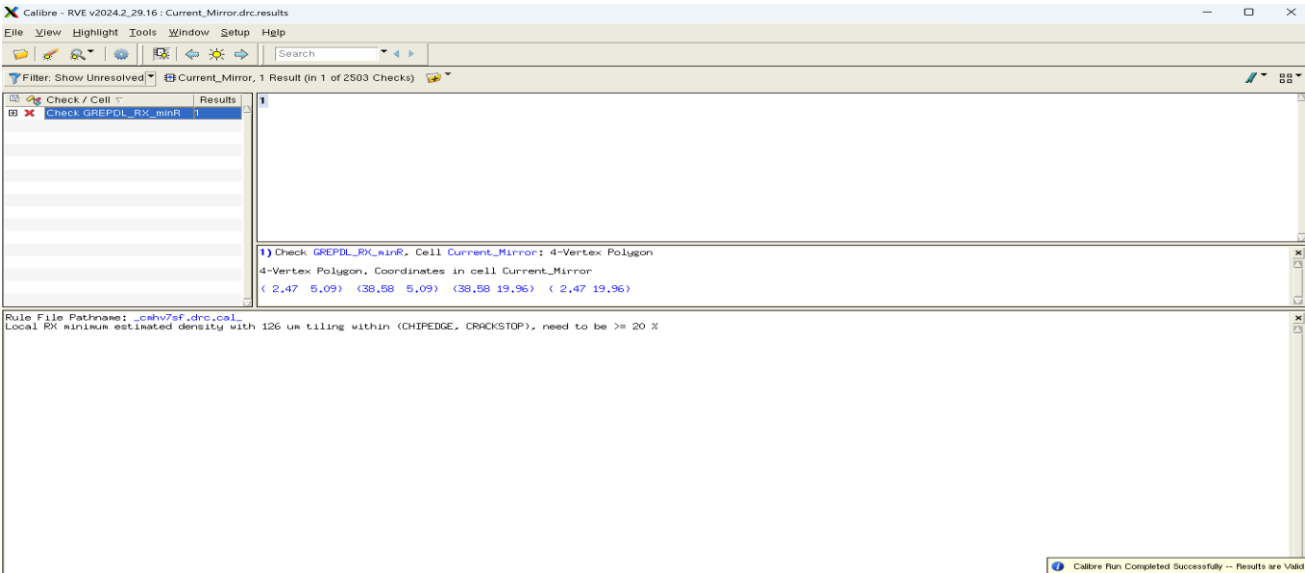


3.Resistor Floor Plan:

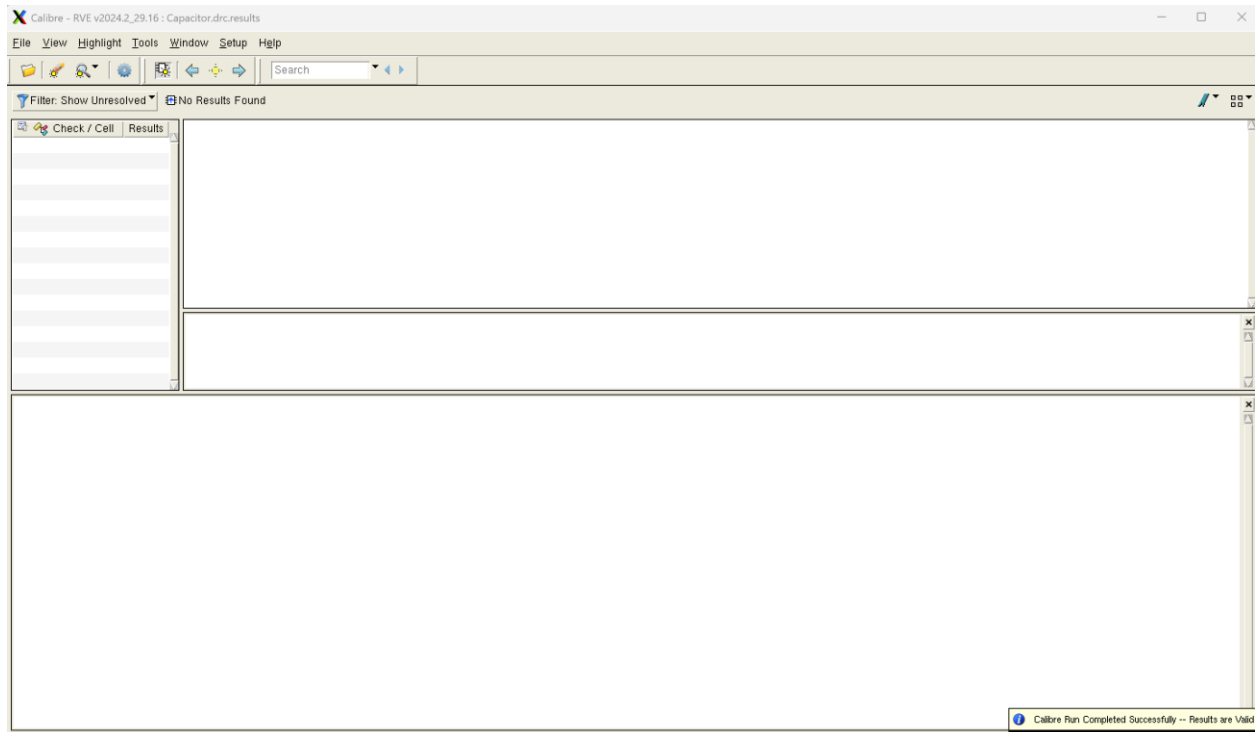


IV. DRC AND LVS RESULTS

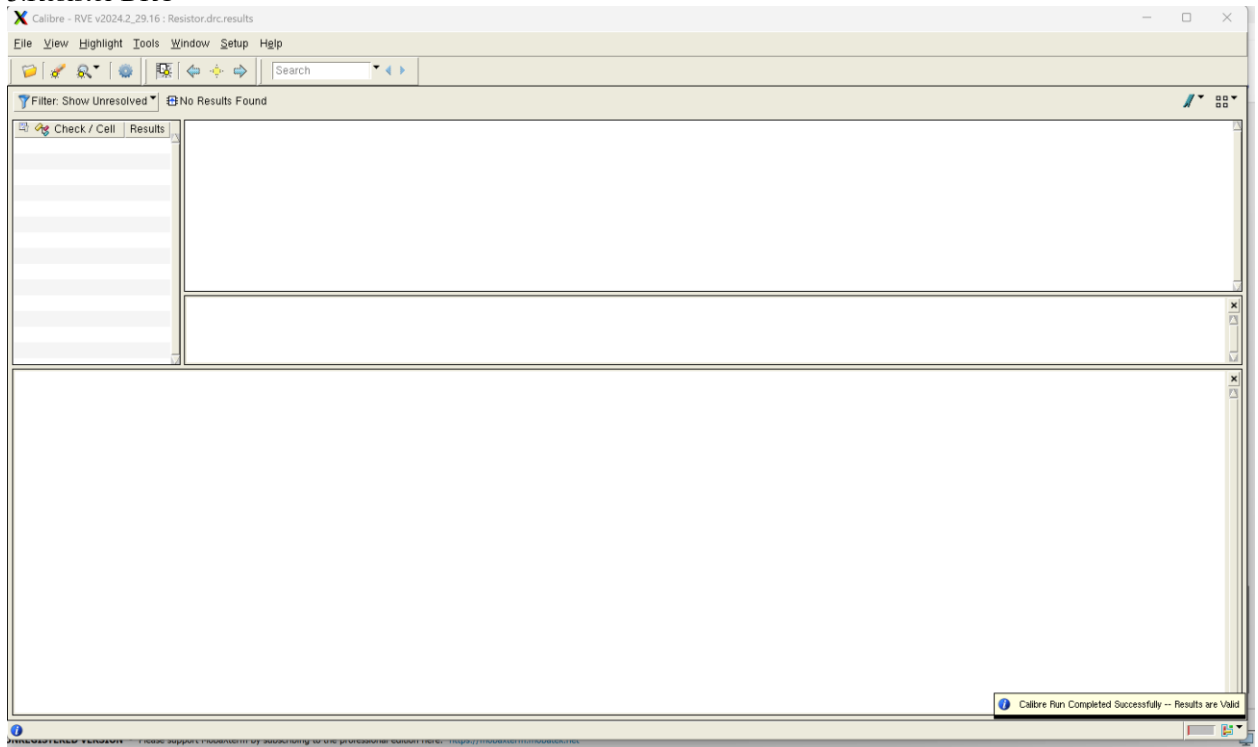
1.Current Mirror DRC



2.Capacitor DRC



3.Resistor DRC



1.Current Mirror LVS

Calibre Interactive - nml2v v2024.2.29.16 /disk/ams/IBM_PDK/cmhu7sf/rel/Calibre/runsetLvs *@n01-zeus.olympus.ece.tamu.edu

File Settings Configurations Help

Rules
Inputs
H-Cells
Outputs
ERC
Signatures
Run Control
Search
Transcript
Files

Current_Mirror.lvs.report

```
11 REPORT FILE NAME: Current_Mirror.lvs.report
12 LAYOUT NAME: Current_Mirror.sp ('Current_Mirror')
13 SOURCE NAME: Current_Mirror.src.net ('Current_Mirror')
14 RULE FILE: _cmhu7sf.lvs.cal
15 CREATION TIME: Tue Oct 7 17:51:46 2025
16 CURRENT DIRECTORY: /home/grads/s/shes123/CAL
17 USER NAME: shes123
18 CALIBRE VERSION: v2024.2_29.16 Thu May 2 07:35:42 PDT 2024
```

OVERALL COMPARISON RESULTS

```
28 #####
29 # ##### - -
30 # # *
31 # # CORRECT # |
32 # # # \_/_/
33 # # #####
34 #####
```

CELL SUMMARY

Result	Layout	Source
CORRECT	Current_Mirror	Current_Mirror

LVS PARAMETERS

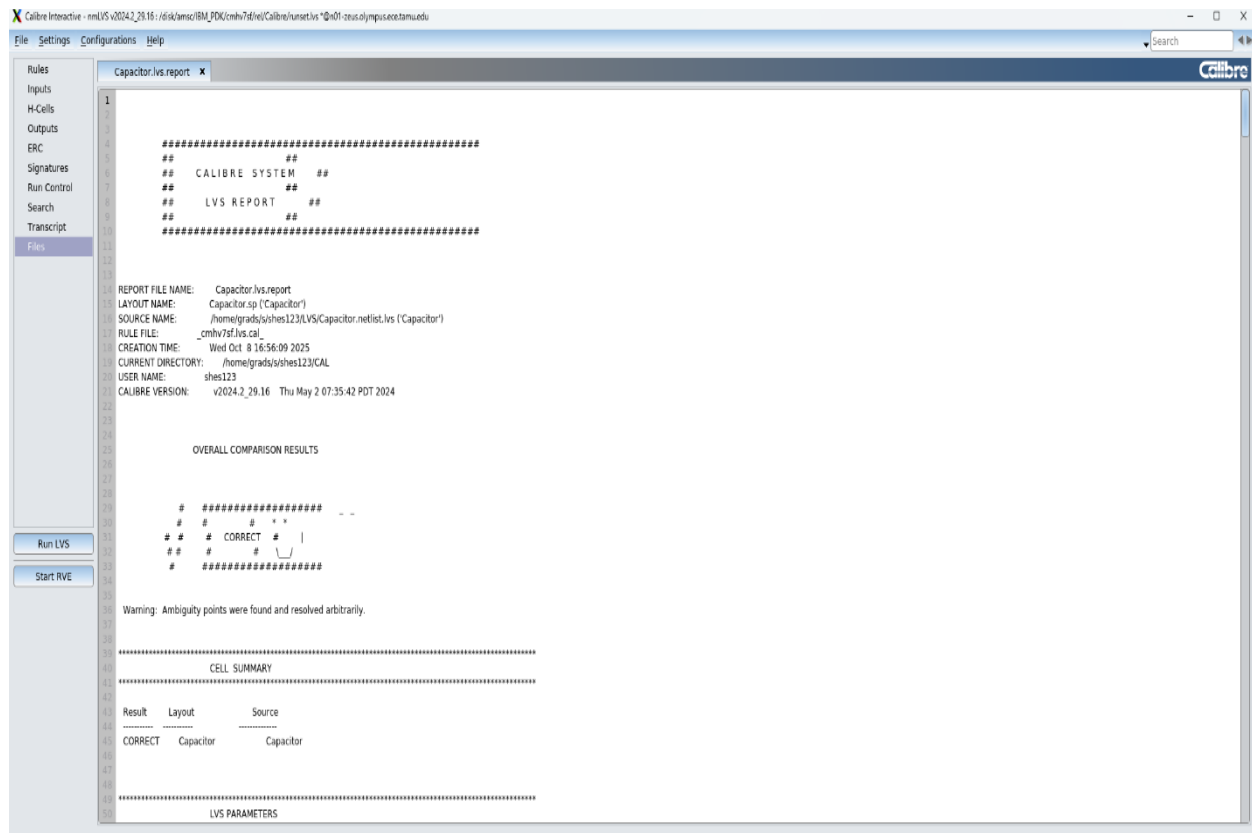
o LVS Setup:

```
54 LVS COMPONENT TYPE PROPERTY element
55 LVS COMPONENT SUBTYPE PROPERTY model
56 LVS PIN NAME PROPERTY phy_pin
57 LVS POWER NAME "Vcc" "vcc" "VCC?"
58 "VDD" "Vdd" "vdd" "VDD:P" "DVDD" "DVDD:P" "VDD!" "TVDD?" "vdd?" "inh_vdd" "INH_VDD" "VCC"
59 "VSS" "Vss" "vss" "VSS:G" "DVSS" "DVSS:P" "TVSS?" "vss?" "GND" "GND:G" "GND!" "TGND?"
60 LVS GROUND NAME "sub!" "inh_gnd" "INH_GND"
61 LVS CELL SUPPLY NO
62 LVS RECOGNIZE GATES ALL
63 // LVS HCELL REPORT
64 LVS IGNORE PORTS NO
65 LVS CHECK PORT NAMES NO
66 LVS IGNORE TRIVIAL NAMED PORTS NO
67 LVS BUILTIN DEVICE PIN SWAP NO
68 LVS ALL CAPACITOR PINS SWAPPABLE NO
69 LVS DISCARD PINS BY DEVICE NO
70 LVS SOFT SUBSTRATE PINS NO
71 LVS INJECT LOGIC YES
72 LVS EXPAND UNBALANCED CELLS YES
73 LVS FLATTEN INSIDE CELL NO
74 LVS EXPAND SEED PROMOTIONS NO
75 LVS PRESERVE PARAMETERIZED CELLS NO
76 LVS GLOBALS ARE PORTS YES
77 LVS RELAX INITIAL CPOINTS NONE
78 LVS REVERSE WL NO
79 // LVS NETLIST FILTER DEVICES NO
80 // LVS PRESERVE BOX PORTS NO
81 LVS SPICE PREFER PINS NO
82 LVS SPICE SLASH IS SPACE YES
83 LVS SPICE ALLOW FLOATING PINS YES
84 // LVS SPICE ALLOW INLINE PARAMETERS
```

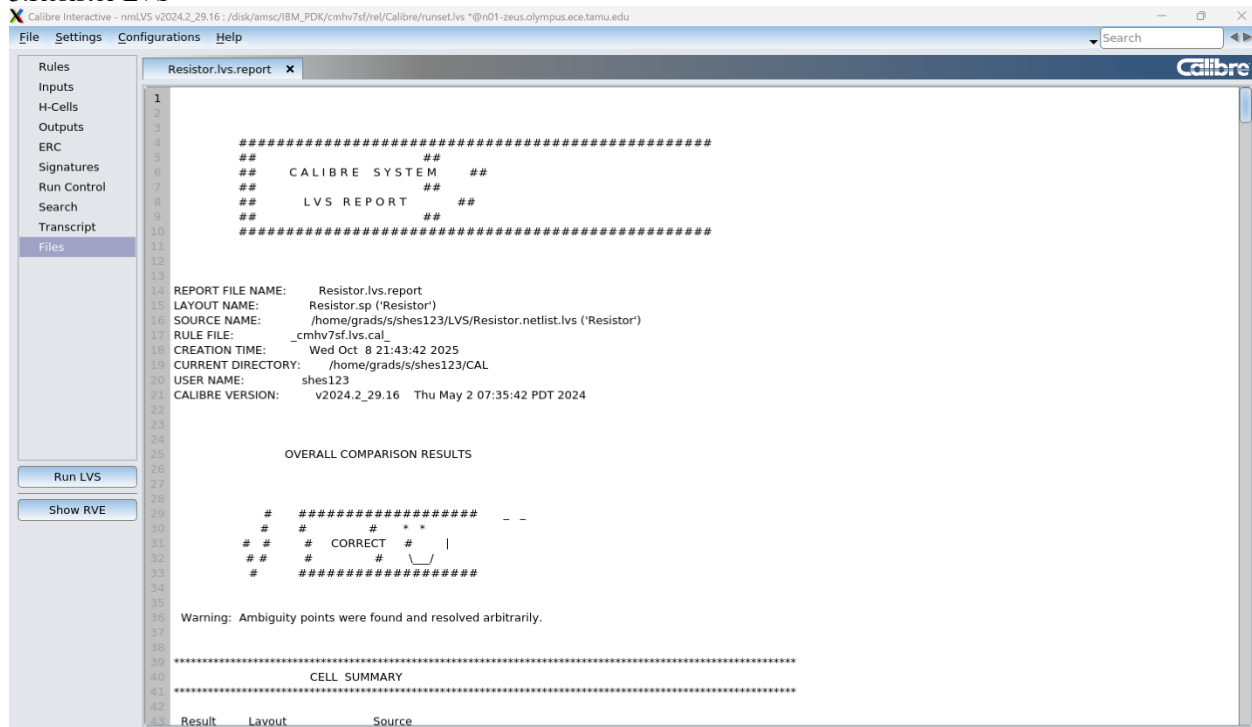
Run LVS

Show RVE

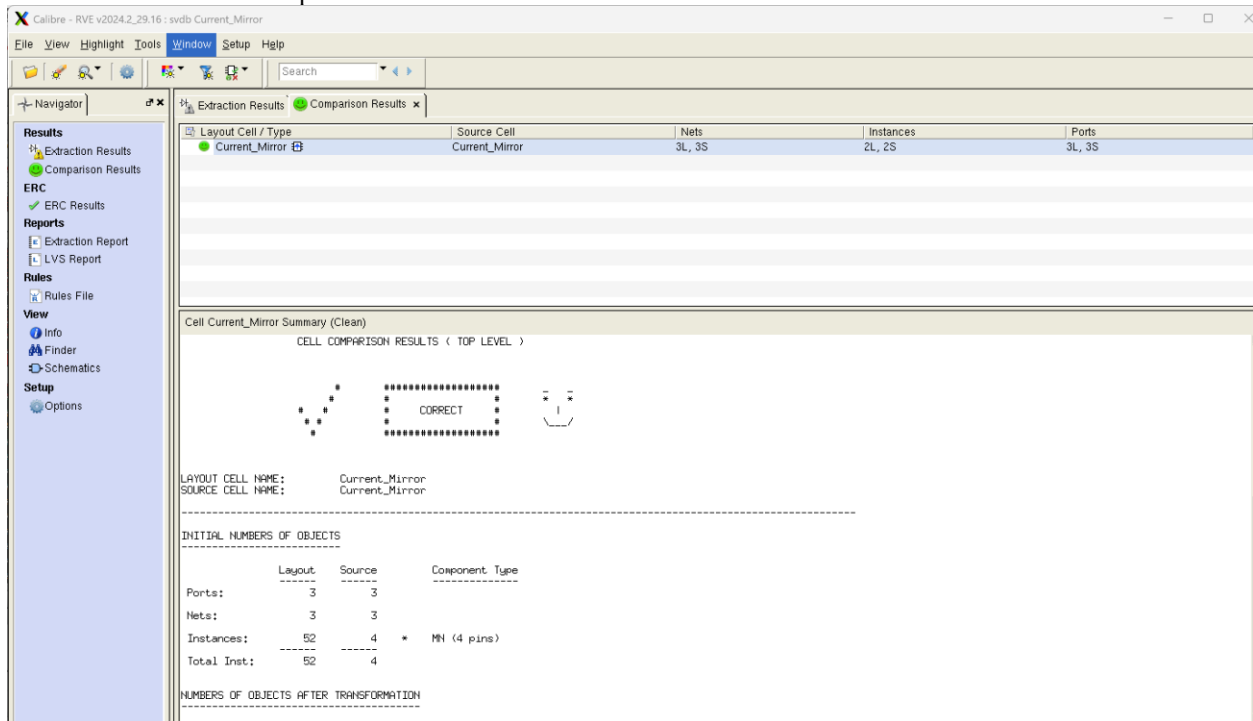
2.Capacitor LVS



3.Resistor LVS



1.Current Mirror LVS Comparison



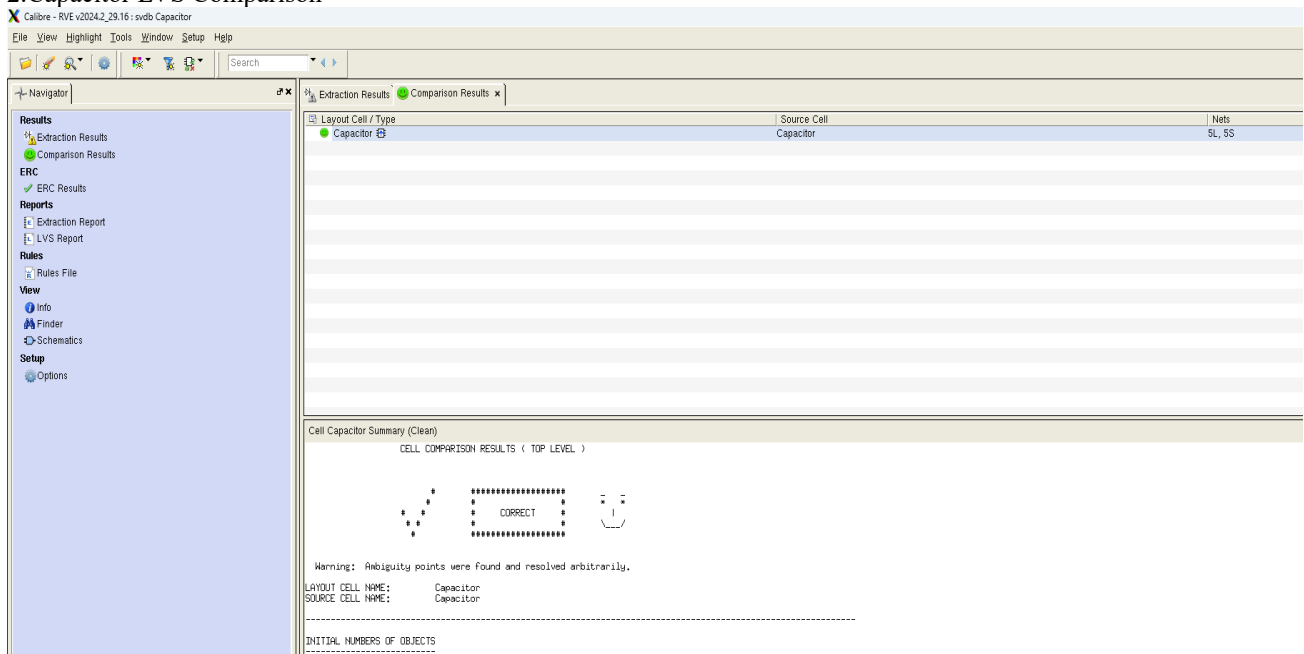
The screenshot shows the Calibre v2024.2.29.16 interface for the 'svdb Current_Mirror' project. The 'Comparison Results' tab is active, displaying a table with the following data:

Layout Cell / Type	Source Cell	Nets	Instances	Ports
Current_Mirror	Current_Mirror	3L, 3S	2L, 2S	3L, 3S

Below the table, the 'Cell Current_Mirror Summary (Clean)' section shows the 'CELL COMPARISON RESULTS (TOP LEVEL)' as 'CORRECT'. It also displays the 'INITIAL NUMBERS OF OBJECTS' and 'NUMBERS OF OBJECTS AFTER TRANSFORMATION'.

	Layout	Source	Component Type
Ports:	3	3	
Nets:	3	3	
Instances:	52	4	* MN (4 pins)
Total Inst:	52	4	

2.Capacitor LVS Comparison



The screenshot shows the Calibre v2024.2.29.16 interface for the 'svdb Capacitor' project. The 'Comparison Results' tab is active, displaying a table with the following data:

Layout Cell / Type	Source Cell	Nets
Capacitor	Capacitor	5L, 5S

Below the table, the 'Cell Capacitor Summary (Clean)' section shows the 'CELL COMPARISON RESULTS (TOP LEVEL)' as 'CORRECT'. It also displays the 'INITIAL NUMBERS OF OBJECTS' and 'NUMBERS OF OBJECTS AFTER TRANSFORMATION'.

Warning: Ambiguity points were found and resolved arbitrarily.

	Layout	Source	Component Type
Ports:	3	3	
Nets:	3	3	
Instances:	52	4	* MN (4 pins)
Total Inst:	52	4	

Calibre - RVE v2024.2_29.16 : svdb Resistor

